AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM359
Systems Penetration Testing

Final Examination
Fall Semester 2024/2025

Date: 04 /Jan/2025

Number of Exam Pages: 2 (including this cover sheet)	**Time Allowed:** 3 H

Instructions:

- Total Marks: **50**
- This exam consists of **five Questions**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Question 1 [10 marks]

For each question fill in the blank space with an appropriate word, acronym, name or phrase. For each blank space you must give only one answer. However, there may be more than one correct answer. Each question is worth 1 mark.

1. ---------- Threat Intelligence type that consumed by security operation center staff and incident response team.
   ---------- is a type of social engineering attack that involves tricking people into giving sensitive information such as login credentials or credit card numbers.
2. ---------- is a technique used to prevent SQL injection attacks by ensuring that user input is validated and sanitized before it is sent to a database. This can prevent malicious code from being injected into a query and executed.
3. ---------- serve as a mean of gaining information about attackers, attack techniques, and tolls by storing a record of the system activities.
4. ---------- programs that hide their presence as well as attacker's activities, granting them full access to the server or host at the time and in the future.
5. ---------- algorithms are used to encrypt and decrypt data. AES, RSA, and SHA are all examples of cryptography algorithms.
6. ---------- is the practice of concealing a message or file within another message or file.
7. ---------- is a technique used by attackers to hide their tracks by erasing evidence of their activities from log files. This can make it more difficult for defenders to detect and respond to the attack.
8. ---------- is a vulnerability in web applications where an attacker can inject malicious code into a web page viewed by other users.
9. The attacker copies the target's password file and then tries to crack passwords in his system at a different location. What type of password attack that performed ---- ----------

Question 2 [10 marks]

1. Explain in detail the process of SNMP, what is the version should be used? (5 marks).
2. Explain in detail the difference between IPS vs IDS and SPAN port? (5 Marks)

Question 3 [10 marks]

1. What is the difference between static malware and dynamic malware analysis? List some static malware analysis techniques. (5 Marks)
2. Explain in detail, OWASP top 10, Broken access and how to avoid its attacks. (5 marks)

Question 4 [10 marks]

1. Explain, Trojan? (3 Marks)
2. What is Steganography? (3 Marks)
3. Briefly explain HTTPS process. (4 Marks)

Question 5 : Explain in detail the following: (10 marks)

1. SQL Injection and what is WAF. (5 Marks)
2. OWASP top 10, give example of 2 vulnerabilities (5 Marks)

End of Questions

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